

Solutions to JEE Advanced Booster Test – 3 | 2022 | Code A

[PHYSICS]

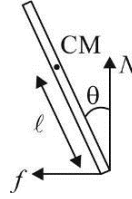
1.(BCD) $N = mg$

$$f = \frac{mv^2}{r}$$

Balancing torque about centre of mass

$$N\ell \sin \theta = f\ell \cos \theta$$

$$\tan \theta = \frac{f}{N} = \frac{mv^2}{rmg} = \frac{v^2}{rg}$$



As mg passes through CM, so the resultant of N and f (which is the net force by the road) should also pass through the CM

2.(ACD) In figure (i), the spring balance measures the tension which is equal to mg

$$\therefore mg = 40$$

In figure (iii), $T' = mg - B$

$$B = \rho vg = \rho \frac{mg}{2\rho} = \frac{mg}{2} = 20$$

$$\therefore T' = 20N$$

In figure (ii), the weighing machine measures the combined weight of (container + water).

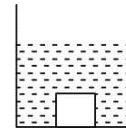
In figure (iii), the (container + water) system experiences a downward force by the block which is reaction to the buoyant force forced by water on the block.

$$\therefore \text{New reading} = 100 + 20 = 120N$$

Note that if you also include block in the system, then buoyant force becomes internal. New reading will be calculated as

$$\begin{aligned} \text{New reading} &= \text{total weight} - T' \\ &= (100 + 40) - 20 = 120N \end{aligned}$$

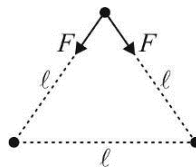
For option (D), taking (container + water + block) as system, the weighing machine will read the total weight i.e. $(100 + 40) = 140N$



3.(BCD) Net force $= 2F \cos 30^\circ$

$$= \sqrt{3}F = \sqrt{3} \frac{Gm^2}{\ell^2}$$

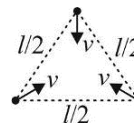
$$\therefore a = \frac{\sqrt{3}Gm}{\ell^2}$$



When separation reduces to $\ell/2$, applying energy conservation

$$0 - \frac{3Gm^2}{\ell} = 3 \frac{1}{2} mv^2 - \frac{3Gm^2}{(\ell/2)}$$

$$\frac{3}{2} mv^2 = \frac{3Gm^2}{\ell} \Rightarrow v = \sqrt{\frac{2Gm}{\ell}}$$



4.(AB) Consider the liquid column in the horizontal portion AB.

Pressure difference will only occur if the acceleration of CM is along AB.

In option B, acceleration is towards B

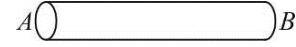
$$\text{So, } P_A > P_B$$

In option D, radial acceleration of CM is towards the axis i.e. A

$$\text{So, } P_B > P_A$$

In option A and C, acceleration is not along AB

$$\text{So, } P_A = P_B$$



5.(ABD) After collision, the velocity required by the combined mass 2m to escape is given by

$$\frac{1}{2} 2mv_e^2 - \frac{GM 2m}{2R} = 0 \Rightarrow v_e = \sqrt{\frac{GM}{R}}$$

Let velocity of satellite A in its orbit be V and velocity of projectile B just before collision be v'.

$$\therefore \left(\frac{v'}{2}\right)^2 + \left(\frac{v}{2}\right)^2 = v_e^2 = \frac{GM}{R}$$

$$\text{Also } v = \sqrt{\frac{GM}{2R}} \quad (\text{orbital velocity})$$

$$\therefore \frac{v'^2}{4} = \frac{GM}{R} - \frac{GM}{8R} = \frac{7}{8} \frac{GM}{R} \quad \therefore v' = \sqrt{\frac{7}{2} \frac{GM}{R}}$$



Energy conservation for B gives

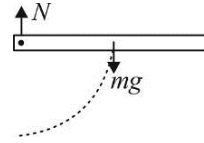
$$\frac{1}{2} mu^2 - \frac{GMm}{R} = \frac{1}{2} mv'^2 - \frac{GMm}{2R}$$

$$\frac{1}{2} mu^2 = \frac{1}{2} m \left(\frac{7}{2} \frac{GM}{R} \right) + \frac{GMm}{2R}; \quad u^2 = \frac{9}{2} \frac{GM}{R} \Rightarrow u = \sqrt{\frac{9GM}{2R}}$$

6.(ABC) Just after release

$$\tau_H = I_H \alpha$$

$$mg \frac{\ell}{2} = \frac{m\ell^2}{3} \alpha \Rightarrow \alpha = \frac{3g}{2\ell}$$



$$\text{Tangential acceleration of CM} = \frac{\alpha \ell}{2} = \frac{3g}{4} \quad \therefore mg - N = m \frac{3g}{4} \Rightarrow N = \frac{mg}{4}$$

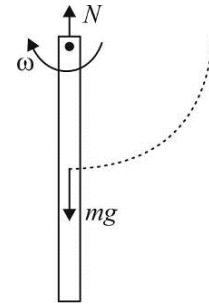
There is no force by hinge in horizontal direction as radial acceleration of CM is zero. (as $\omega = 0$)

When the rod becomes vertical, its angular velocity is given by

$$mg \frac{\ell}{2} = \frac{1}{2} \frac{m\ell^2}{3} \omega^2 \Rightarrow \omega = \sqrt{\frac{3g}{\ell}}$$

$$\text{Radial acceleration of CM} = \omega^2 \frac{\ell}{2} = \frac{3g}{2}$$

$$\therefore N - mg = m \frac{3g}{2} \Rightarrow N = \frac{5}{2} mg$$



There is no force by hinge in horizontal direction as tangential acceleration of CM is zero (as $\tau_H = 0$)

7.(C) Applying Bernoulli between points just inside and just outside the hole

$$P_1 + \frac{1}{2} 2\rho v_1^2 = P_2 + \frac{1}{2} 2\rho v_2^2; \quad P_0 + \rho g \frac{H}{2} + 2\rho g \left(\frac{H}{2} - h \right) + 0 = P_0 + \frac{1}{2} 2\rho v^2$$

$$v^2 = g \frac{H}{2} + gH - 2gh; \quad v = \sqrt{g(1.5H - 2h)}$$

8.(C) $T = \sqrt{\frac{2h}{g}}$

$$R = vT = \sqrt{\frac{2h}{g}} g(1.5H - 2h) = \sqrt{3hH - 4h^2}$$

For R to be maximum $\frac{d}{dh}(3hH - 4h^2) = 0 \Rightarrow 3H - 8h = 0 \Rightarrow h = \frac{3H}{8}$

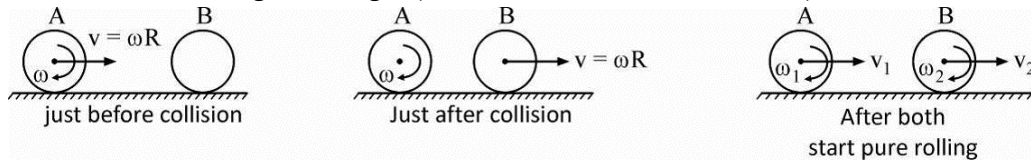
9.(B) 10.(C)

During collision, no impulsive friction force acts on A or B.

$$\therefore J_{CM} = 0 \Rightarrow \Delta L_{CM} = 0$$

So, there is no change in angular velocity and just after collision $\omega_A = \omega$ & $\omega_B = 0$.

Their linear velocities get exchanged (same mass elastic head on collision).



After both start pure rolling :

$$\text{For A : } \frac{2}{5}mR^2\omega = \frac{2}{5}mR^2\omega_1 + mV_1R \Rightarrow \frac{2}{5}mR^2\omega = \frac{2}{5}mR^2\left(\frac{V_1}{R}\right) + mV_1R \Rightarrow V_1 = \frac{2}{7}\omega R$$

$$\text{For B : } mVR = \frac{2}{5}mR^2\omega_2 + mV_2R \Rightarrow m(\omega R)R = \frac{2}{5}mR^2\left(\frac{V_2}{R}\right) + mV_2R \Rightarrow V_2 = \frac{5}{7}\omega R$$

11.(B) (i) Time period, $T = 2\pi\sqrt{\frac{r^3}{GM}}$

$$\text{Since } M = \rho\left(\frac{4}{3}\pi R^3\right); T \propto \sqrt{\frac{a^3}{\rho R^3}}$$

(ii) Force, $F = \frac{GMm}{r^2}$ Therefore, $F \propto \frac{\rho R^3 m}{a^2}$

(iii),(iv) Kinetic energy, $K = \frac{GMm}{2r}$ Total energy, $T.E. = -\frac{GMm}{2r}$

Therefore, both K and T.E. are proportional to $\frac{\rho R^3 m}{a}$

12.(C) Taking clockwise torque as positive, and anti-clockwise torque as negative

Torque on the body due to weight of OA, OB and OC (about O) is

$$\tau_{OA} = (2mg)\left(\frac{L}{2}\cos\theta\right) = \frac{4}{5}mgL$$

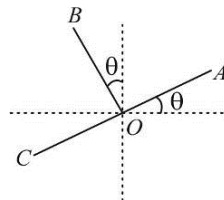
$$\tau_{OB} = -(mg)\left(\frac{L}{2}\sin\theta\right) = -\frac{3}{10}mgL$$

$$\tau_{OC} = -(mg)\left(\frac{L}{2}\cos\theta\right) = -\frac{2}{5}mgL$$

So, net torque due to gravity about O,

$$\tau_O = \tau_{OA} + \tau_{OB} + \tau_{OC} = \frac{1}{10}mgL$$

So, net torque due to gravity is $\frac{1}{10}mgL$, clockwise



Therefore, torque due to F must be $\frac{1}{10}mgL$ anti-clockwise to maintain equilibrium.

(i) F is applied horizontally at A

$$\tau_F = F(L \sin \theta) = \frac{1}{10}mgL \Rightarrow F = \frac{1}{6}mg \quad (\text{towards left})$$

(ii) F is applied vertically at B

$$\tau_F = F(L \sin \theta) = \frac{1}{10}mgL \Rightarrow F = \frac{1}{6}mg \quad (\text{downwards})$$

(iii) F is applied perpendicular to OC , at mid-point of OC

$$\tau_F = F\left(\frac{L}{2}\right) = \frac{1}{10}mgL \Rightarrow F = \frac{1}{5}mg$$

(iv) F is applied vertically at C

$$\tau_F = F(L \cos \theta) = \frac{1}{10}mgL \Rightarrow F = \frac{1}{8}mg \quad (\text{downwards})$$

13.(0.75) Above the surface,

$$g' = \frac{GM}{(R+h)^2} = \frac{GM}{R^2} \left(\frac{R}{R+h}\right)^2 = g \left(\frac{R}{R+h}\right)^2$$

$$\frac{g}{4} = g \left(\frac{R}{R+h}\right)^2 \Rightarrow \frac{1}{2} = \frac{R}{R+h} \Rightarrow h = R$$

Below the surface, $g' = \frac{GM}{R^2} \left(1 - \frac{d}{R}\right) = g \left(1 - \frac{d}{R}\right)$

$$\frac{g}{4} = g \left(1 - \frac{d}{R}\right) \Rightarrow \frac{1}{4} = 1 - \frac{d}{R} \Rightarrow d = \frac{3R}{4} \quad \therefore \frac{d}{h} = \frac{3}{4} = 0.75$$

14.(13.33)

The vertical forces acting on the apparatus is its weight and the forces due to atmosphere pressure and liquid pressure as shown.

$$F_{atm} = P_0 3\pi R^2, \text{ where } R = 10 \text{ cm}$$

$$F_{liquid} = [P_0 + \rho g(h - h_0)] 3\pi R^2$$

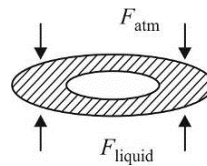
The system lifts up when

$$F_{liquid} - F_{atm} = mg$$

$$\rho g(h - h_0) 3\pi R^2 = mg$$

$$1000(h - h_0) 3(10^{-2}) = 1$$

$$h - h_0 = \frac{1}{30} \text{ m} = 3.33 \text{ cm} ; h = 13.33 \text{ cm}$$

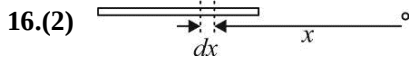


15.(0.75) $a = \frac{g \sin \theta}{1 + \frac{k^2}{R^2}}$; where k is radius of gyration

$$\text{For hollow cylinder, } mR^2 = mk^2 \Rightarrow \frac{k^2}{R^2} = 1 \quad \therefore a_H = \frac{g \sin \theta}{2}$$

$$\text{For solid cylinder, } \frac{mR^2}{2} = mk^2 \Rightarrow \frac{k^2}{R^2} = \frac{1}{2}$$

$$\therefore a_s = \frac{g \sin \theta}{1 + \frac{1}{2}} = \frac{2}{3} g \sin \theta ; \frac{a_H}{a_s} = \frac{3}{4} = 0.75$$



Mass of element of length dx is $dM = \frac{M}{\ell} dx$

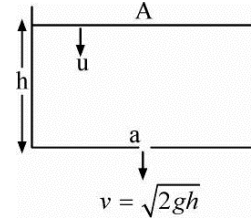
Force between the particle and this element is $dF = \frac{Gm(dM)}{x^2} = \frac{GMm}{\ell} \frac{dx}{x^2}$

So total force $= \int dF = \frac{GMm}{\ell} \int_{\ell}^{2\ell} \frac{dx}{x^2} = \frac{GMm}{\ell} \left[-\frac{1}{x} \right]_{\ell}^{2\ell} = \frac{GMm}{2\ell^2}$

17.(2) $Au = a\sqrt{2gh}$

$$\Rightarrow A \left(-\frac{dh}{dt} \right) = a\sqrt{2gh} \Rightarrow \int_{H_1}^{H_2} -\frac{dh}{\sqrt{h}} = \int_0^t \frac{a}{A} \sqrt{2g} dt$$

$$\Rightarrow -\left[2\sqrt{h} \right]_{H_1}^{H_2} = \frac{a}{A} \sqrt{2g} t \Rightarrow t = \frac{A}{a} \sqrt{\frac{2}{g}} [\sqrt{H_1} - \sqrt{H_2}]$$



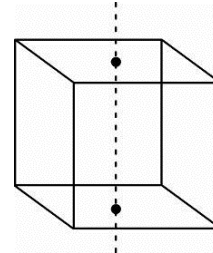
Now, $T_1 = \frac{A}{a} \sqrt{\frac{2}{g}} \left[\sqrt{H} - \sqrt{\frac{H}{9}} \right]$ and $T_2 = \frac{A}{a} \sqrt{\frac{2}{g}} \left[\sqrt{\frac{H}{9}} - \sqrt{0} \right] \therefore \frac{T_1}{T_2} = \frac{2}{3} \sqrt{H} / \frac{1}{3} \sqrt{H} = 2$

18.(0.28) Mass of each face $= \frac{M}{6}$

M.I. of top and bottom faces are equal and is given by $I = \left(\frac{M}{6} \right) \frac{a^2}{6} = \frac{Ma^2}{36}$

M.I. of remaining four faces are equal and is given by $I' = \left(\frac{M}{6} \right) \frac{a^2}{12} + \frac{M}{6} \left(\frac{a}{2} \right)^2 = \frac{Ma^2}{18}$

$$\therefore I_{\text{box}} = 2I + 4I' = \frac{2Ma^2}{36} + \frac{4Ma^2}{18} = \frac{5Ma^2}{18}$$



- 1.(A) Use $\Delta S = nC_p \ln(T_2 / T_1) + nR \ln(P_1 / P_2)$
- 2.(ACD) Adding more solid will not shift the equilibrium because solids are not part of equilibrium expressions. Neither will adding inert gas into a container with a constant volume because it is not part of the reaction. While A may look correct, it will only speed up the reaction, thus decreasing time to reach equilibrium. But this process does not change the equilibrium state itself. Since the reaction is endothermic, decreasing the temperature will favor the reverse exothermic reaction by Le Chatelier's principle. Hence answer is ACD.
- 3.(ABCD) Entropy increase with increase in number of particles.
- 4.(ABCD) At 1 atm and 100.0°C, water reaches its boiling point. This means that it can turn into vapour if heat is added, or it can turn into liquid by releasing heat. Thus, the vaporization is reversible at 1 atm and 100.0°C. Now let's think about the phase diagram of water. If we increase the pressure past 1 atm, water becomes a liquid, and the vaporization is no longer spontaneous. During vaporization entropy increases
Order of strength of H-bond, Solid > Liquid > Gas
- 5.(ABCD) Conversion of thermodynamically less stable substance to thermodynamically more stable substance is exothermic for example conversion of diamond to graphite is exothermic, P(white, solid) to P(Black, solid) is exothermic.
As $\Delta S_{\text{surrounding}} = -\Delta S_{\text{system}}$ (for Reversible process)
 $\Delta S_{\text{system}} = \Delta H_{\text{sys}} / T$, As $\Delta S_{\text{Surr}} = (+)\text{ve}$, $\therefore \Delta S_{\text{sys}} = (-)\text{ve}$, $\Delta H_{\text{sys}} = (-)\text{ve}$
- 6.(D) Every system has a tendency to attain a state of lowest free energy.
At equilibrium state Gibbs Free energy of the system is minimum
- 7.(B) ΔH° and ΔS° are temperature independent.
- 8.(B) For Sn(white $\rightarrow$ Sn(gray), $\Delta H^\circ < 0$ and $\Delta S^\circ < 0$ hence reaction is spontaneous at $T < T_{\text{switch}}$. $T_{\text{switch}} = (\Delta H^\circ / \Delta S^\circ) = 13.78^\circ\text{C}$.
At temperature lower than 13.78°C forward reaction is spontaneous i.e. gray tin is stable and at temperature higher than 13.78°C backward reaction is spontaneous i.e. white tin is stable.
- 9.(C) On increasing T, endothermic reaction shift in forward direction and on addition of inert gas(Ar) at constant pressure equilibrium shift to greater number of moles.
- 10.(A) The decrease in volume will lead to an increase in pressure. According to Le Chatelier's principle, the equilibrium is going to shift to the left side with less moles of gases as pressure increases. This is because the shift to fewer gas moles will reduce the pressure increase. First, let's find K_p using the equilibrium data.

$$K_p = \frac{(P_{\text{SO}_2})(P_{\text{Cl}_2})}{(P_{\text{SO}_2\text{Cl}_2})} = \frac{(0.10)(0.10)}{(1.00)} = 0.010$$

Now let's find Q_p . Since the volume was halved, all pressures are now doubled.

$$Q_p = \frac{(P_{\text{SO}_2})(P_{\text{Cl}_2})}{(P_{\text{SO}_2\text{Cl}_2})} = \frac{(0.20)(0.20)}{(2.00)} = 0.020$$

This confirms that the reaction must shift left. As the reaction reaches equilibrium, the pressure of SO_2Cl_2 will increase by some amount x . We see from the reaction that SO_2 and Cl_2 will decrease by the same amount. Thus, $K_p = \frac{(0.20-x)(0.20-x)}{(2.00+x)}$

We could solve this equation for x to find the final pressure of SO_2Cl_2 , but it would take time. Instead, notice that $x > 0$ because by Le Chatelier's principle, SO_2Cl_2 must increase. Also notice that $x < 0.20$ because otherwise the partial pressures of SO_2 and Cl_2 would be negative. Therefore, the final SO_2Cl_2 is between 2.00 bar and 2.20 bar, which fits the range of an increase 1-10%. Thus, the answer is A. If we were to actually solve for x , we would get 0.06, meaning an increase of 3%.

11.(B)

	Reaction	Change in Thermodynamics Function
I.	$3\text{O}_2 \longrightarrow 2\text{O}_3(\text{g})$	Always non-spontaneous $\Delta_r H > 0$ and $\Delta_r S < 0$, $\Delta_r G > 0$, $\Delta S_{\text{uni}} < 0$
II.	$2\text{C}(\text{Solid, graphite}) + \text{O}_2(\text{g}) \longrightarrow 2\text{CO}(\text{g})$	$\Delta_r H < 0$ and $\Delta_r S > 0$, $\Delta_r G < 0$ (always)
III.	$\text{P}(\text{Solid, white}) \longrightarrow \text{P}(\text{Solid, Red})$	$\Delta_r H < 0$ and $\Delta_r S < 0$, $\Delta_r G < 0$ only at low temperature
IV.	$\text{NH}_4\text{Cl}(\text{s}) \longrightarrow \text{NH}_3(\text{g}) + \text{HCl}(\text{g})$	$\Delta_r H > 0$ and $\Delta_r S > 0$, $\Delta_r G < 0$ only at high temperature

12.(A)

	Process	Variable
I.	Isothermal expansion of ideal gas	$\Delta U = 0$, $w = -q$
II.	Free expansion	$w = 0$, $q = 0$, $\Delta U = 0$
III.	Adiabatic expansion	$q = 0$, $w = \Delta U$
IV.	Isochoric	$w = 0$, $q = \Delta U$

13.(10) We can set up the following equation, writing the concentration as $\frac{n}{V}$ where n is the number of moles.

$$K_C = \frac{[\text{Cd}((\text{SCN})_4)^{2-}]}{[\text{Cd}^{2+}][\text{SCN}^-]^4} = \frac{\left(\frac{1.0}{V}\right)^1}{\left(\frac{1.0}{V}\right)^5}; \quad 1.0 \times 10^4 = V^4 \times (1.0)^{-4}$$

Solving, we get $V = 10 \text{ L}$

14.(67.24) K_p for the first reaction is $K_p = \frac{(P_{\text{SO}_2})(P_{\text{O}_2})^{1/2}}{(P_{\text{SO}_3})}$

$$K_p \text{ for the new reaction is } K_p = \frac{(P_{\text{SO}_2})^2(P_{\text{O}_2})}{(P_{\text{SO}_3})^2}$$

We see that since all coefficients were multiplied by 2, the new K_p is equal to the old one squared.

$$(8.2 \times 10^{-4})^2 = 67.24 \times 10^{-8}$$

- 15.(-1615.10) Let x be the standard enthalpy of formation of $\text{SiF}_4(\text{g})$. Using products minus reactants from the reaction and the table, we have $(x - 2 \cdot 285.8) - (-910.9 - 4 \cdot 320.1) = 4.6$

Solving for x we get : $-1615.1 \text{ kJ mol}^{-1}$

- 16.(0.08) $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$

$$\begin{array}{ccc} 0.05 & 0.05 & 0 \\ 0.05 - x & 0.05 - x & 2x \end{array}$$

$$50.41 = \frac{4x^2}{(0.05 - x)^2}$$

$$7.1 = \frac{2x}{(0.05 - x)}$$

$$0.355 - 7.1x = 2x$$

$$0.355 = 9.1x$$

$$x = \frac{0.355}{9.1} = 0.0390$$

$$2x = 0.078 = 0.08$$

- 17.(-328) $\text{HO}^- + \text{HCO}_3^- \longrightarrow \text{CO}_3^{2-} + \text{H}_2\text{O}$

$$\begin{array}{ccc} \frac{0.2 \times 40}{1000} & \frac{0.1 \times 200}{1000} & - \quad - \\ 0.008 & 0.02 & \end{array}$$

$$0 \quad (0.02 - 0.008) \quad 0.008 \quad 0.008$$

$$\Delta_r H = [(-677 \times 0.008) + (-286 \times 0.008)] - [(-692 \times 0.008) + (-230 \times 0.008)]$$

$$= (-963 \times 0.008) - (-922 \times 0.008)$$

$$= -41 \times 0.008 = -328 \text{ J}$$

- 18.(18) $\text{H}_2\text{O}_{(\text{s})} \xrightarrow{+q_1} \text{H}_2\text{O}_{(\ell)} \xrightarrow{+q_2} \text{H}_2\text{O}_{(\ell)} \xleftarrow{-q_3} \text{H}_2\text{O}_{(\ell)}$

$$\begin{array}{cccc} 0^\circ\text{C} & 0^\circ\text{C} & t^\circ\text{C} & 32^\circ\text{C} \\ 50\text{ gm} & 50\text{ gm} & & 350\text{ gm} \end{array}$$

Heat absorbed ($q_1 + q_2$) = Heat evolved (q_3)

$$(50 \times 334.72) + (4.184 \times 50 \times (t - 0)) = 4.184 \times 350 \times (32 - t)$$

$$334.72 \times 50 = 4.184 \times 350 \times (32 - t) - 4.184 \times 50 \times (t - 0)$$

$$= 4.184 \times 50 [7(32 - t) - t]$$

$$\frac{334.72 \times 50}{4.184 \times 50} = [224 - 8t]$$

$$80 = 224 - 8t$$

$$8t = 224 - 80$$

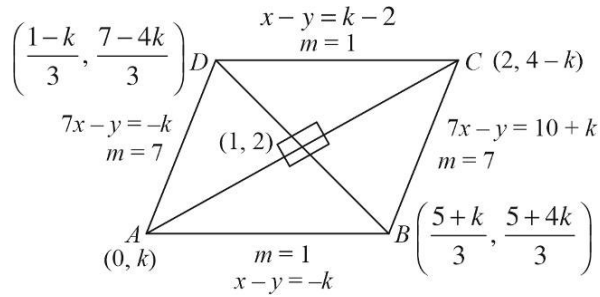
$$t = 18^\circ\text{C}$$

1.(AD) (0, 0) & A(6, 3) lies on opposite side of BC i.e. $x + y = 2$

Orthocenter is point of intersection of $x - y = 3$ and $2x + 5y = 19$ i.e. $H\left(\frac{34}{7}, \frac{13}{7}\right)$

H lies inside $\triangle ABC$

2.(ABCD)



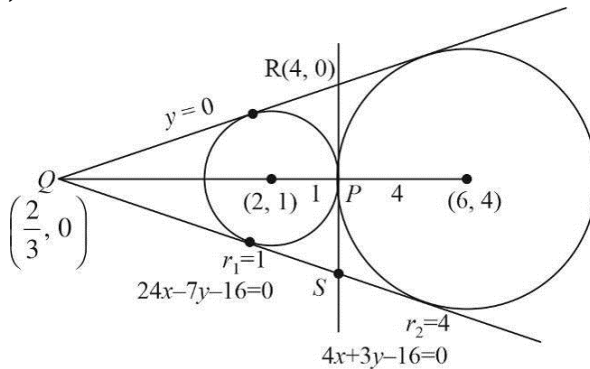
$$\left(\frac{2-k}{1}\right)\left(\frac{1-4k}{-2-k}\right) = -1$$

$$2 - 8k - k + 4k^2 = 2 + k$$

$$4k^2 - 10k = 0$$

$$k = 0, \frac{5}{2}$$

3.(ABD)



$$\text{Point of contact } P\left(\frac{4 \times 2 + 1 \times 6}{4 + 1}, \frac{4 \times 1 + 1 \times 4}{4 + 1}\right) = \left(\frac{14}{5}, \frac{8}{5}\right)$$

Equation of common tangent at point of contact is $8x + 6y - 32 = 0$ i.e. $4x + 3y - 16 = 0$

Let $y = m\left(x - \frac{2}{3}\right)$ be common tangent

$$\left| \frac{m\left(2 - \frac{2}{3}\right) - 1}{\sqrt{m^2 + 1}} \right| = 1$$

$$\left(\frac{4}{3}m - 1\right)^2 = m^2 + 1$$

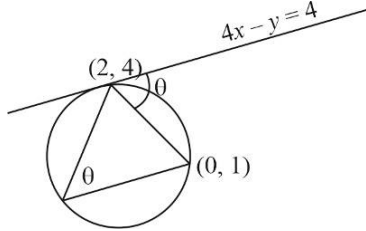
$$7m^2 - 24m = 0$$

$$m=0, \frac{24}{7} \Rightarrow y=0 \text{ \& } 24x-7y-16=0$$

$$Q\left(\frac{2}{3}, 0\right), R(4, 0), S\left(\frac{8}{5}, \frac{16}{5}\right)$$

NOTE: area of Δ formed by the three-common tangent to two circles touching each other externally is equal to $\frac{2(r_1 r_2)^{3/2}}{|r_1 - r_2|}$

4.(AC)



$$\text{Let circle be } (x-2)^2 + (y-4)^2 + \lambda(4x-y-4) = 0$$

$$x^2 + y^2 + 4(\lambda-1)x - (\lambda+8)y + 20 - 4\lambda = 0$$

Let it pass through (0, 1)

$$\Rightarrow 1 - \lambda - 8 + 20 - 4\lambda = 0$$

$$\lambda = \frac{13}{5} \Rightarrow x^2 + y^2 + \frac{32}{5}x - \frac{53}{5}y + \frac{48}{5} = 0$$

$$\text{Centre } \left(-\frac{16}{5}, \frac{53}{10}\right) \quad \tan \theta = \frac{4 - \frac{3}{2}}{1 + 4 \cdot \frac{3}{2}} = \frac{5}{14}$$

$$\begin{aligned} 5.(\text{ABCD}) \quad \sum_{r=1}^{20} A_r B_r &= \sum_{r=1}^{20} {}^{20}C_r {}^{40}C_r = {}^{20}C_1 \times {}^{40}C_1 + {}^{20}C_2 \times {}^{40}C_2 + \dots + {}^{20}C_{20} \times {}^{40}C_{20} \\ &= {}^{60}C_{40} - 1 \text{ or } {}^{60}C_{20} - 1 = C_{40} - 1 \text{ or } C_{20} - 1 \end{aligned}$$

$$\sum_{r=1}^{20} A_r^2 = \sum_{r=1}^{20} ({}^{20}C_r)^2 = ({}^{20}C_1)^2 + ({}^{20}C_2)^2 + \dots + ({}^{20}C_{20})^2 = {}^{40}C_{20} - 1 = B_{20} - 1$$

$$\begin{aligned} \sum_{r=0}^{20} A_r C_r &= \sum_{r=0}^{20} {}^{20}C_r \cdot {}^{60}C_r = {}^{20}C_0 \cdot {}^{60}C_0 + {}^{20}C_1 {}^{60}C_1 + \dots + {}^{20}C_{20} \times {}^{60}C_{20} \\ &= {}^{80}C_{20} \end{aligned}$$

$$\sum_{r=0}^{20} (A_r + B_r) = \sum_{r=0}^{20} ({}^{20}C_r + {}^{40}C_r) = 2^{20} + \frac{2^{40} + {}^{40}C_{20}}{2} = 2^{20} + 2^{39} + {}^{39}C_{20}$$

$$6.(\text{AD}) \quad (1-2\sqrt{x})^{50} = 1 - {}^{50}C_1(2\sqrt{x})^1 + {}^{50}C_2(2\sqrt{x})^2 - {}^{50}C_3(2\sqrt{x})^3 + \dots + {}^{50}C_{50}(2\sqrt{x})^{50}$$

$$(1+2\sqrt{x})^{50} = 1 + {}^{50}C_1(2\sqrt{x})^1 + {}^{50}C_2(2\sqrt{x})^2 + \dots + {}^{50}C_{50}(2\sqrt{x})^{50}$$

$$(1-2\sqrt{x})^{50} + (1+2\sqrt{x})^{50} = 2\left(1 + {}^{50}C_2(2\sqrt{x})^2 + \dots + {}^{50}C_{50}(2\sqrt{x})^{50}\right)$$

$$= 2\left(1 + {}^{50}C_2 2^2 x + {}^{50}C_4 2^4 x^2 + \dots + {}^{50}C_{50} \cdot 2^{50} x^{25}\right)$$

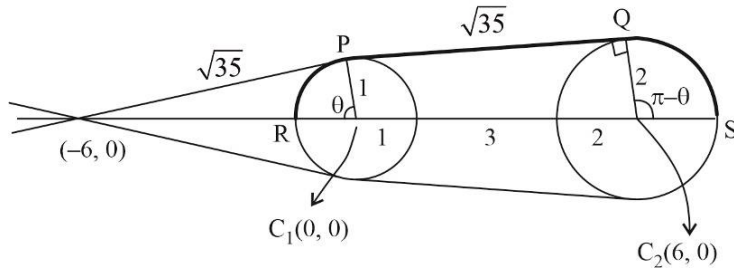
$$\text{Let } x=1 \Rightarrow 1 + {}^{50}C_2 2^2 + {}^{50}C_4 2^4 + \dots + {}^{50}C_{50} 2^{50} = \frac{3^{50} + 1}{2}$$

$$\frac{(4-1)^{50} + 1}{2} = \frac{4k+1+1}{2} = 2k+1$$

$$\text{Similarly, } 3^{50} = (10-1)^{25}$$

Using this option (D) is correct

7.(B), 8.(A)

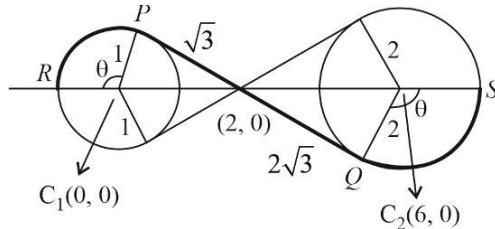


Using $l = r\theta$

$$\widehat{PR} = 1 \times \theta = \theta$$

$$\widehat{QS} = 2(\pi - \theta) = 2\pi - 2\theta$$

$$\begin{aligned} \text{Length of belt 1} &= 2(\widehat{PR} + \sqrt{35} + \widehat{QS}) \\ &= 2(2\pi - \theta + \sqrt{35}) \\ &= 2(\sqrt{35} + 2\pi - \tan^{-1} \sqrt{35}) \end{aligned}$$



$$\widehat{PR} = 1 \left(\pi - \frac{\pi}{3} \right) = \frac{2\pi}{3}$$

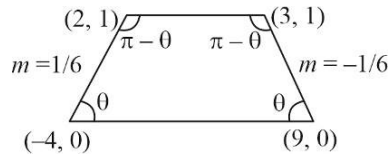
$$\widehat{QS} = 2 \left(\pi - \frac{\pi}{3} \right) = \frac{4\pi}{3}$$

$$\text{Length of belt 2} = 2(\widehat{PR} + \widehat{QS} + PQ) = 2(2\pi + 3\sqrt{3}) = 6\sqrt{3} + 4\pi$$

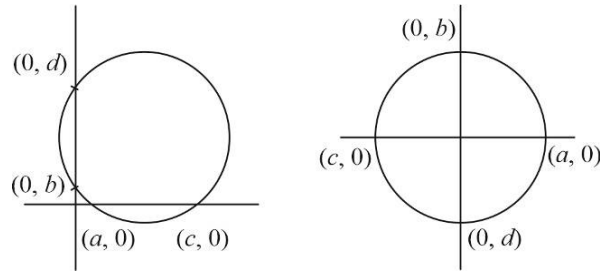
$$\begin{aligned} \text{9.(C)} \quad & \sum_{r=1}^{20} (-1)^{r-1} (r^{20} C_r)^2 \\ &= \sum_{r=1}^{20} (-1)^{r-1} (20 \times {}^{19}C_{r-1})^2 = 400 \sum_{r=1}^{20} (-1)^{r-1} ({}^{19}C_{r-1})^2 \\ &= 400 \left(({}^{19}C_0)^2 - ({}^{19}C_1)^2 + ({}^{19}C_2)^2 - \dots - ({}^{19}C_{19})^2 \right) = 0 \end{aligned}$$

$$\begin{aligned} \text{10.(A)} \quad & \sum_{r=0}^{10} (-1)^r \frac{{}^{10}C_r}{{}^{r+3}C_r} \\ &= \sum_{r=0}^{10} (-1)^r \frac{10!}{r! (10-r)!} \times \frac{r! 3!}{(r+3)!} \\ &= \sum_{r=0}^{10} (-1)^r \frac{10! 3!}{13!} \times \frac{13!}{(r+3)! (10-r)!} \\ &= \frac{1}{13 \times 11 \times 2} \sum_{r=0}^{10} (-1)^r {}^{13}C_{r+3} = \frac{1}{13 \times 11 \times 2} [{}^{13}C_0 - {}^{13}C_1 + {}^{13}C_2] = \frac{1}{13 \times 11 \times 2} [1 - 13 + 78] = \frac{3}{13} \end{aligned}$$

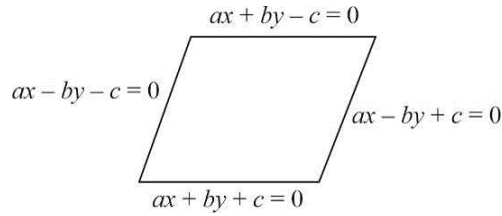
11.(C) (I)



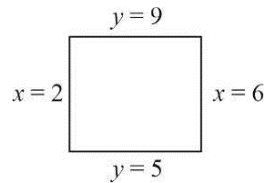
(II)



(III)



(IV)



12.(C) (I)

$$T_{r+1} = {}^{15}C_r (r)^{15-r} \left(\sqrt[3]{3}\right)^r$$

$$r = 0, 1, \dots, 15$$

$$r = 3, 9, 15$$

$$\text{Number of irrational terms} = 16 - 3 = 13$$

$$k = 6$$

$$I \rightarrow p$$

(II)

$$(1+3)^n - 1 = 4095 \Rightarrow n = 6$$

$$II \rightarrow r, s$$

(III)

$${}^nP_r = {}^nC_r \cdot 5040 \Rightarrow r! = 5040 \Rightarrow r = 7$$

$$III \rightarrow s$$

(IV)

$$\sum_{r=1}^{10} r(r-1) {}^{10}C_r = 10 \times 9 \sum_{r=2}^{10} {}^8C_{r-2} = 10 \times 9 \times 2^8$$

$$IV \rightarrow q, r, s$$

$$k = 5$$

13.(1.58)

$$T_{r+1} = {}^{10}C_r \left(\frac{x}{2}\right)^{10-r} \left(-\frac{3}{x^2}\right)^r$$

$$10 - r - 2r = 4$$

$$\Rightarrow r = 2$$

$${}^{10}C_2 \frac{9}{2^8} = \frac{405}{256}$$

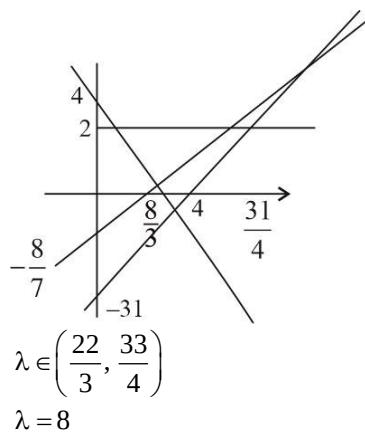
14.(202) $\left((1+x)(1-x+x^2)\right)^{100} (1+x)$

$$(1+x^3)^{100}(1+x); \quad (1+x^3)^{100} + x(1+x^3)^{99}$$

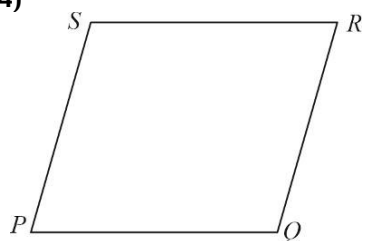
$$= (1 + {}^{100}C_1 x^3 + {}^{100}C_2 x^6 + \dots + {}^{100}C_{100} x^{300}) + (x + {}^{100}C_1 x^4 + {}^{100}C_2 x^7 + \dots + {}^{100}C_{100} x^{301})$$

$$\Rightarrow 101 + 101 = 202 \text{ terms}$$

15.(8)

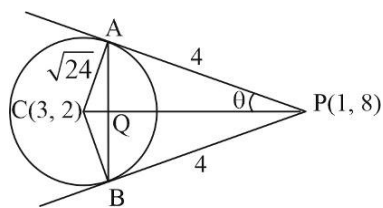


16.(2.24)



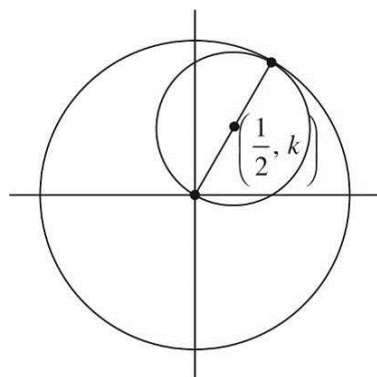
$$\frac{P+R}{2} = \frac{Q+S}{2} \Rightarrow 1+5 = 4+a \Rightarrow a=2 \quad \text{and} \quad 2+7 = 6+b \Rightarrow b=3$$

17.(3.20)



$$\begin{aligned} \text{Area of } \triangle PAB &= 2 \times \frac{1}{2} \times PA \cdot PQ \cdot \sin \theta \\ &= PA \times PA \sin \theta \cdot \cos \theta = 4^2 \times \frac{\sqrt{24}}{\sqrt{40}} \times \frac{4}{\sqrt{40}} = \frac{16}{5} \sqrt{6} \end{aligned}$$

18.(3.75)



$$2\sqrt{\frac{1}{4} + k^2} = 4$$

$$k^2 = 4 - \frac{1}{4} = \frac{15}{4}$$